



Project Title -seasonal. agriculture performance

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PROBLEM STATEMENT

Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyse the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description

This project is to agricultural data from different seasons and identify meaningful patterns, trends, relationships and differences in agricultural performance.

Students should independently:

- ☐ Explore and understand the dataset.
- ☐ Clean and prepare the data for analysis.
- ☐ Examine how agricultural performance varies across seasons.
- ☐ Identify important seasonal patterns and trends.
- ☐ Investigate relationships between seasonal conditions and agricultural outcomes.
- ☐ Compare relevant groups within different seasons.
- ☐ Identify significant differences or unusual patterns.
- ☐ Apply appropriate statistical and graphical techniques.
- ☐ Interpret findings based on evidence from the dataset.
- ☐ Develop meaningful conclusions and data-driven recommendations.

WHO ARE THE END USERS?



For your Seasonal Agriculture Performance Analysis project, the end users are the people or Agencies who can use the analysis and insights to make better agricultural decisions. Your project focuses on seasonal patterns, agricultural performance, resource usage, environmental conditions, and economic outcomes.

1. Farmers
2. Agricultural Officers
3. Agricultural Researchers
4. Government & Agricultural
5. Agribusiness Companies
6. Agricultural Consultants
7. Irrigation & Resource Management Teams
8. Students & Data Analysts



Technology Used

Python – Main programming language for data analysis.

Pandas – Data cleaning, manipulation, and analysis.

NumPy – Numerical calculations and statistical operations.

Matplotlib.pyplot – Creating graphs and visualizations.

Seaborn – Advanced statistical visualizations and charts.

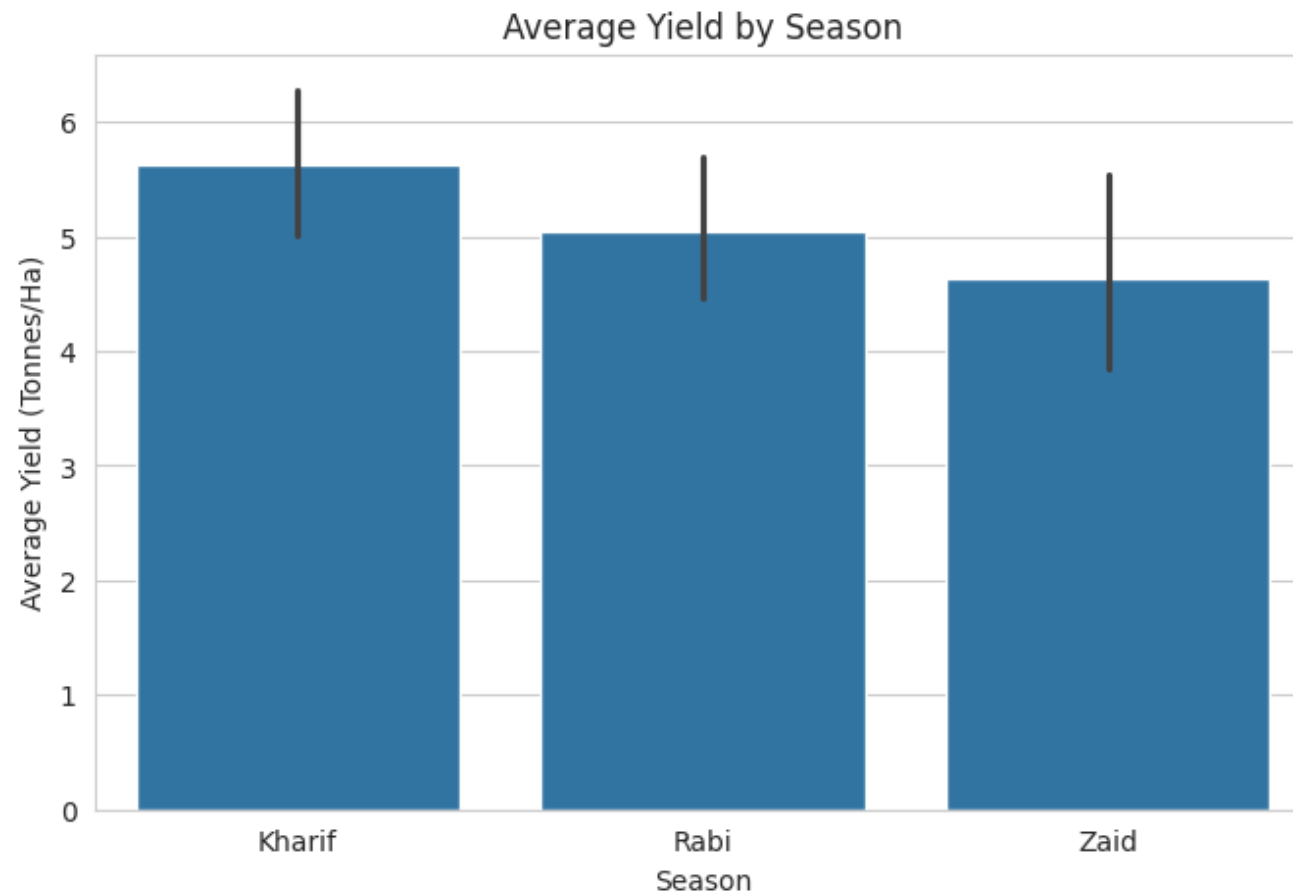
Jupyter Notebook / Google Colab – Environment for writing and executing Python code.

CSV Dataset – Used as the primary data source.

Statistical Analysis – Used to identify relationships, trends, correlations, and seasonal variations.

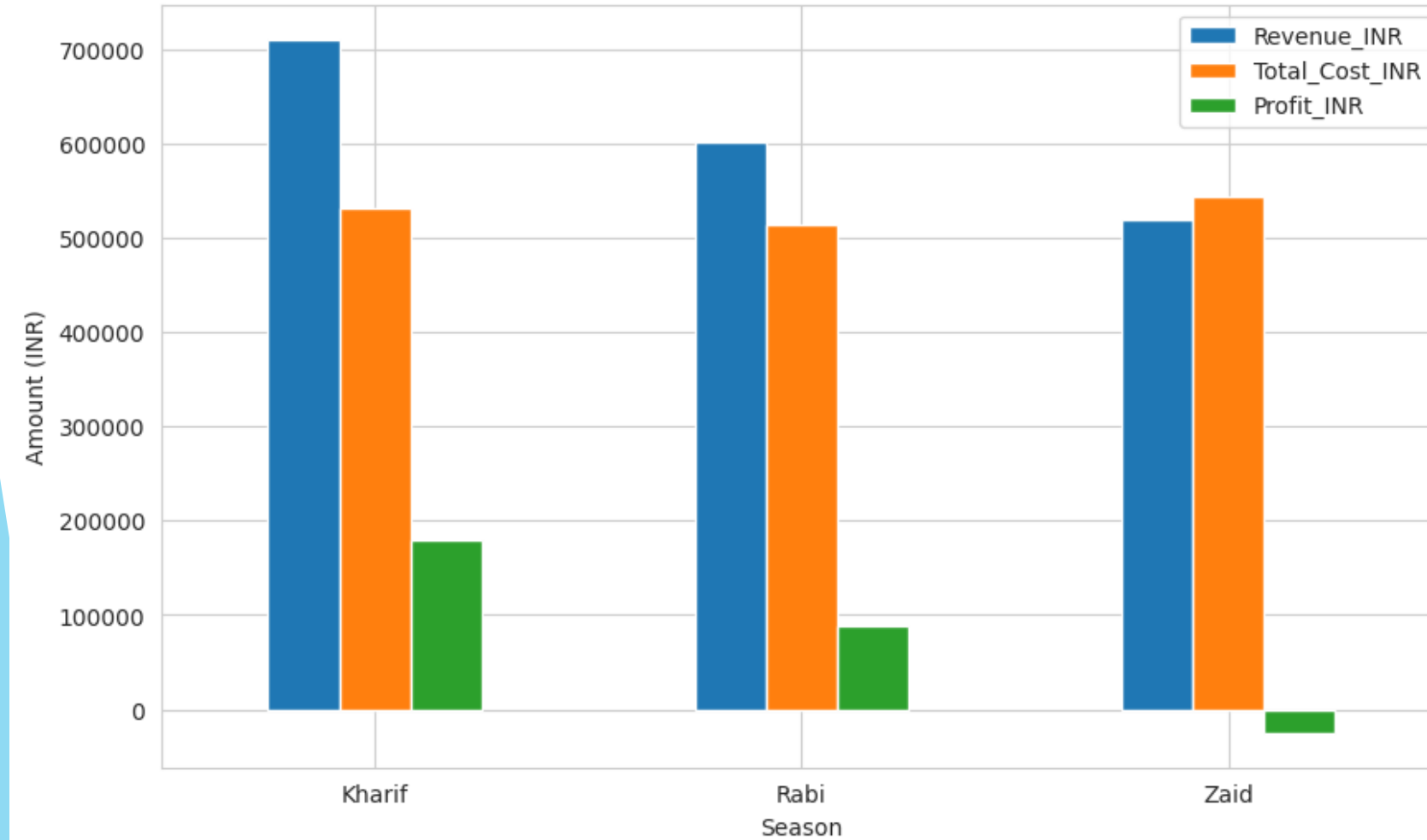


RESULTS

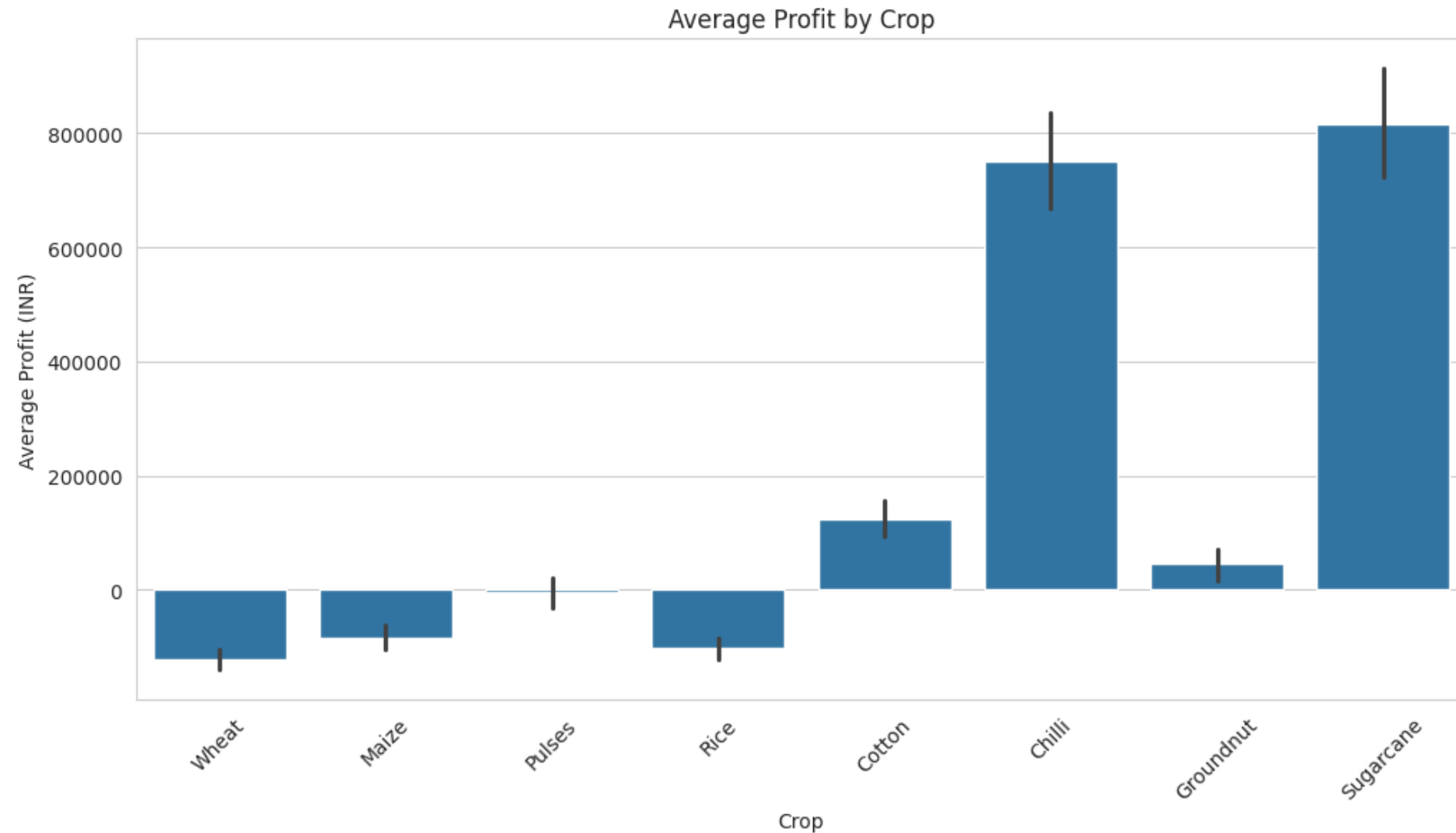


RESULTS

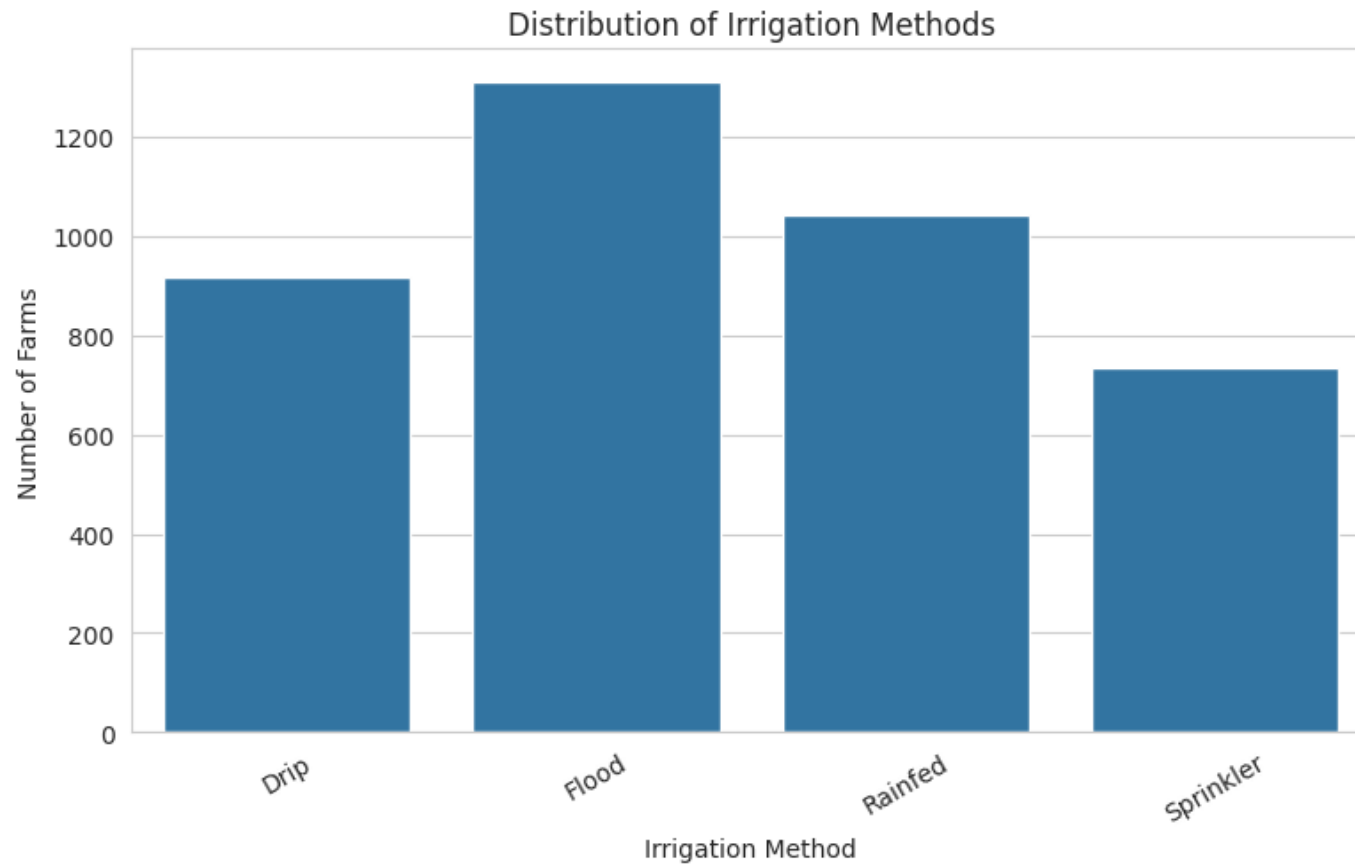
Average Revenue, Cost and Profit by Season



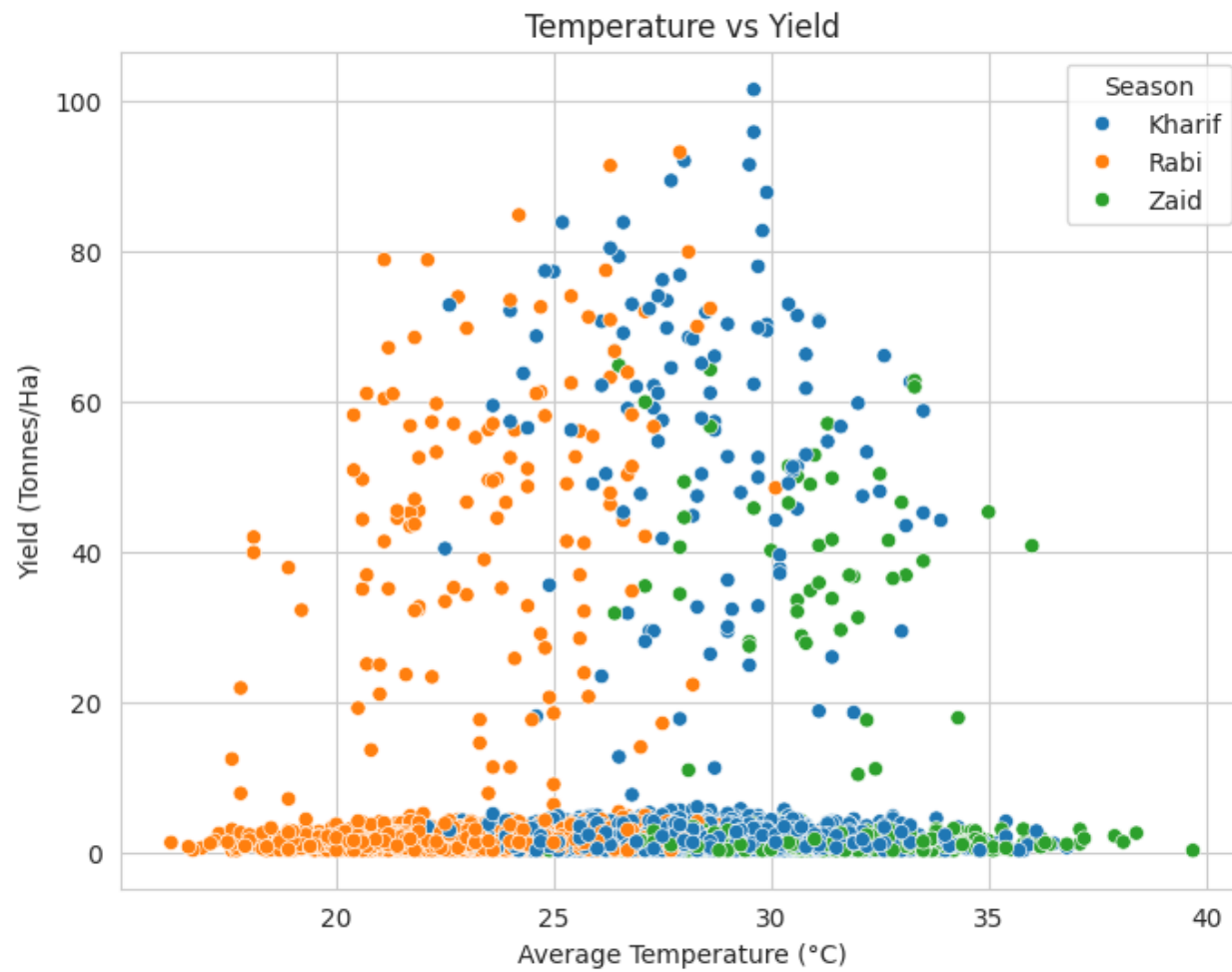
RESULTS



RESULTS



RESULTS



Future scope

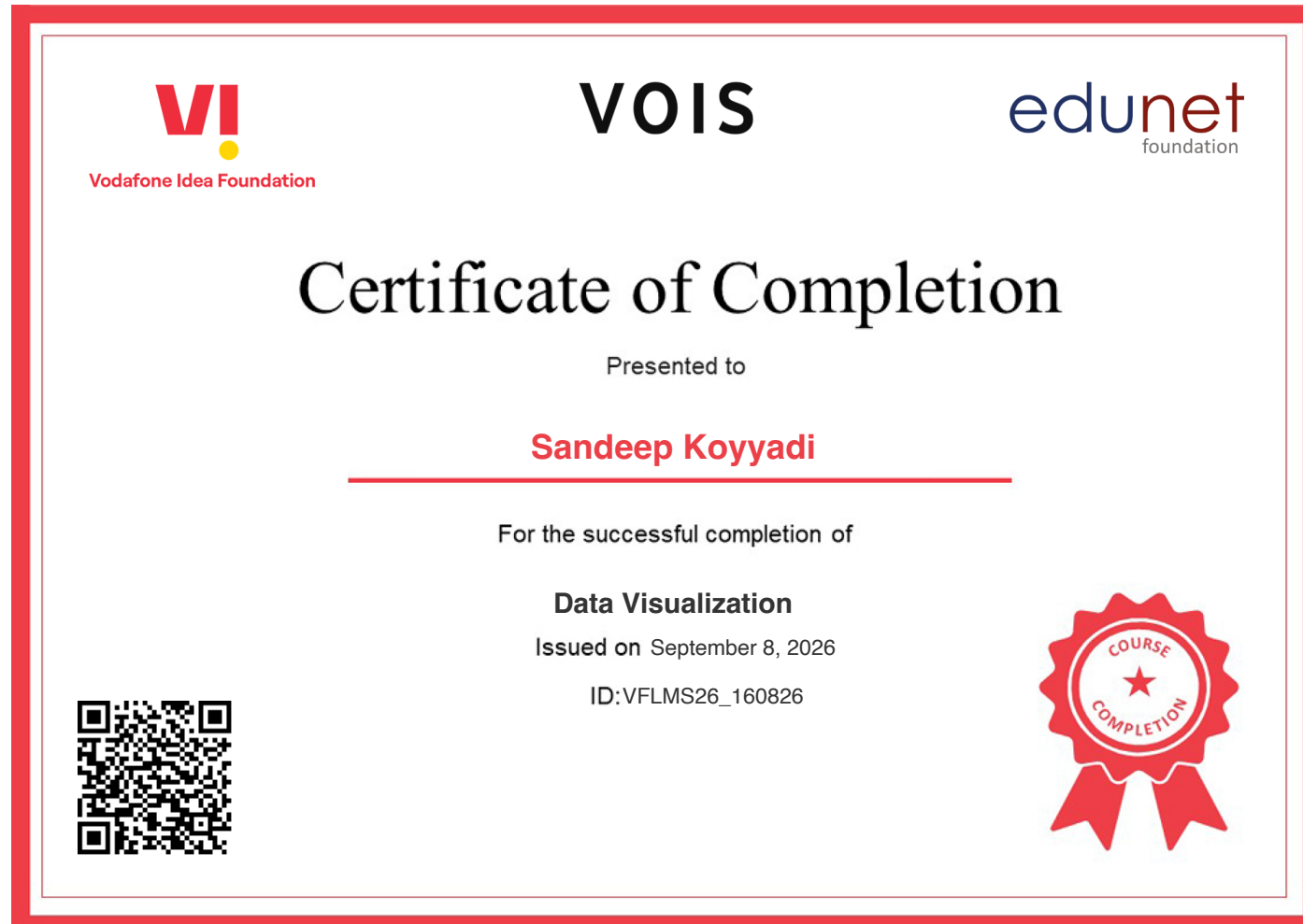
The Seasonal Agriculture Performance Analysis project can be further improved and expanded in the following ways:

1. Real-Time Data Integration
2. Crop Yield Prediction
3. Profit Prediction
4. Weather Forecast Integration
5. Smart Irrigation Analysis
6. Disease & Pest Prediction
7. Interactive Power BI Dashboard
8. Regional Analysis
9. Mobile Application
10. AI-Based Recommendations

GitHub Link

<https://github.com/AswiniKumar55/-Seasonal-Agriculture-Performance-Analysis-.git>

VOIS Course completion certificate - Course name- Data Visualization



Thank you